

Smart Health Risk Predictor — Project Report

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Project Report — Smart Health Risk Predictor

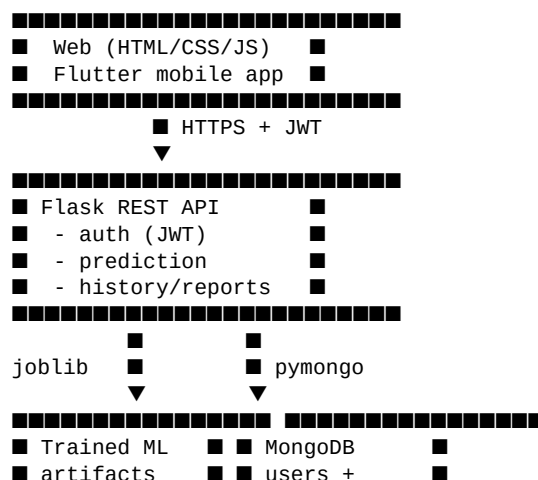
1. Abstract

The Smart Health Risk Predictor is a full-stack application that estimates a user's risk of three common conditions — heart disease, type-II diabetes, and obesity — using supervised machine learning on publicly available clinical datasets. The system exposes its inference via a Flask REST API, provides a responsive HTML/CSS/JavaScript dashboard, and ships a cross-platform Flutter mobile client.

2. Objectives

- Train and compare three classical ML algorithms (Random Forest, Logistic Regression, SVM) on each dataset.
- Persist the best-performing model per disease and serve it through a documented REST API.
- Provide a secure, user-authenticated dashboard that visualises risk probabilities and historical predictions.
- Demonstrate end-to-end production practices: hashed passwords, JWT auth, environment-based config, MongoDB persistence, and cloud deployment.

3. System Architecture



- Dark healthcare theme, fully responsive (CSS grid + media queries).
- Chart .js doughnut & bar visualisations.
- BMI calculator built into the overview tab.

- Forms dynamically generated from `js/predict.js` field definitions so schema changes only need one update.

8. Mobile App

Flutter app reuses the same REST endpoints, persists the JWT with `shared_preferences`, and presents Material 3 dark theme screens for login, registration, dashboard, and per-disease prediction.

9. Security

- Passwords hashed with `bcrypt` (default cost factor 12).
- JWT secret loaded from `JWT_SECRET_KEY` env var.
- Server-side validation on every input (`utils/validators.py`).
- CORS configurable per environment.
- MongoDB Atlas IP allow-list in production.

10. Deployment

Frontend → Vercel · Backend → Render/Railway · Database → MongoDB Atlas.

Full step-by-step in `docs/DEPLOYMENT.md`.

11. Results (sample run)

Numbers depend on dataset version; representative values:

Disease	Best model	Test accuracy	CV accuracy
Heart	Random Forest	~0.98	~0.97
Diabetes	Random Forest	~0.78	~0.76
Obesity	Random Forest	~0.96	~0.95

12. Future Work

- XGBoost / LightGBM benchmarks.
- SHAP-based per-prediction explanations in the dashboard.
- HL7 / FHIR integration for clinical interoperability.
- Background email reports via Celery + Redis.